

Route lifetime vs prediction horizon for three planners

Prediction shortens routes for every planner (Voronoi 518 → 275 ms, PRM 560 → 289 ms, Visibility Graph 341 → 228 ms, 0 → 100 ms); even at 100 ms, event routes last 2.0–2.5× as long as cycle routes without prediction. Projected obstacles block the route sooner, so it is replaced more often. Bars: event (ours) at 20 ms checks; ◇ markers: cycle at 20 ms checks without prediction, for reference. Horizon = how far ahead (ms) obstacles are projected; 0 = no prediction.

